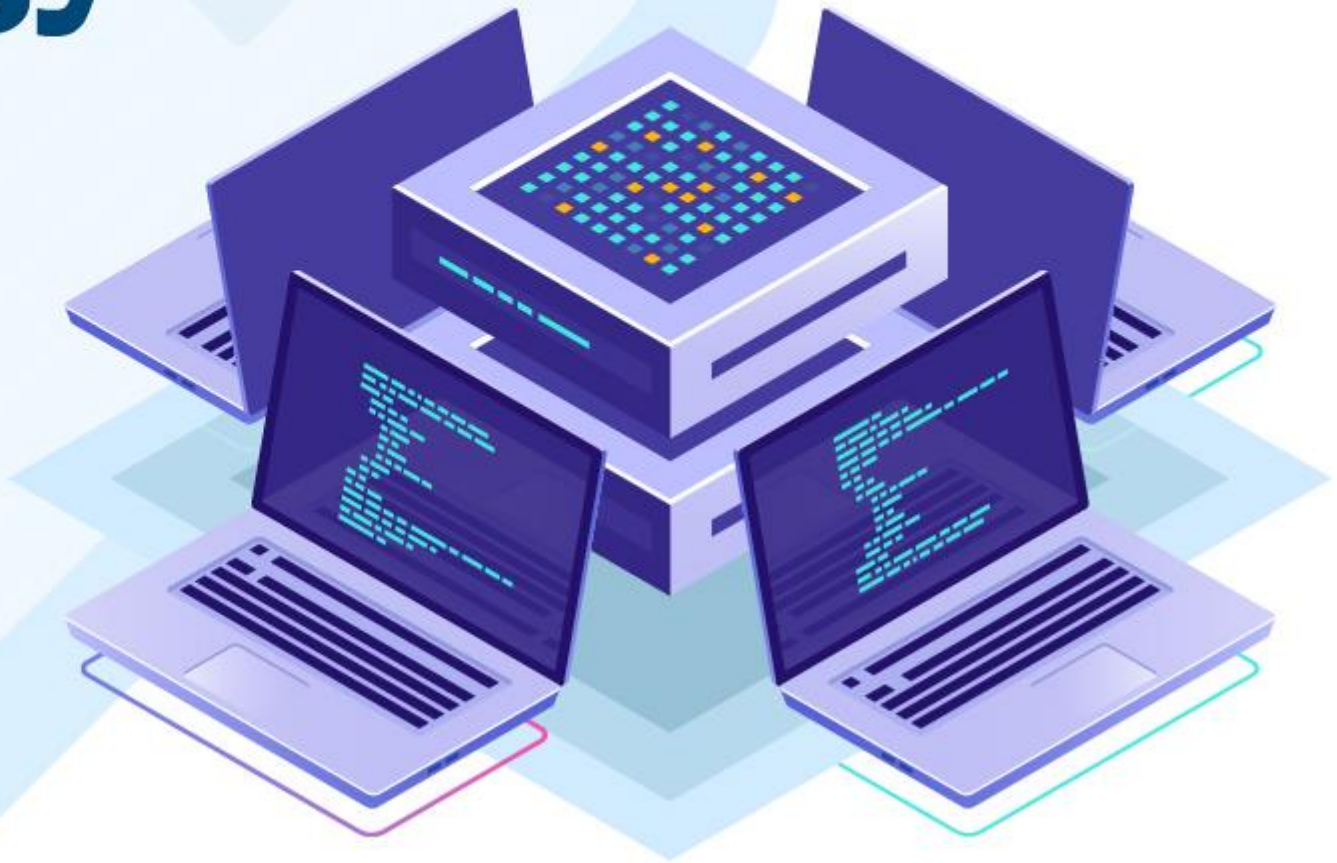
Information Technology

Fall Semester
2026 - 2027

- Embedded Systems
- **Dr. Yasmine Mahmoud**



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Manufacturing Cost & NRE

1- Smart Thermometer:

- A company develops a **smart thermometer**. The company spends $10,000 on hardware design, firmware development, and testing. Each thermometer costs $20 to manufacture.
- What is the NRE?
- What is the manufacturing cost of 500 thermometers?
- What is the total cost?

Manufacturing Cost & NRE

Smart Thermometer:

Answer:

- NRE = $10,000
- Manufacturing cost = 500 × $20 = $10,000
- Total cost =

= NRE + Manufacturing cost

= $10,000 + $10,000

= $20,000

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Manufacturing Cost & NRE

2- A company spends $30,000 developing a **smart door lock**. This includes PCB (Printed Circuit Board) design, firmware, prototypes, and testing. Each door lock costs $50 to manufacture.

The company produces 1,000 units.

What is the total project cost?

Manufacturing Cost & NRE

2- Answer:

NRE = $30,000

Manufacturing cost= 1,000 × $50 = $50,000

Total cost = $30,000 + $50,000 = $80,000

Answer: $80,000

Manufacturing Cost & NRE

3- Increasing Production:

An embedded device has an NRE of $20,000 and a manufacturing cost of $40 per unit. The company produces 100 units.

What is the effective cost per unit?

Answer:

- **NRE per unit: $20,000 ÷ 100 = $200**
- **Manufacturing cost per unit = $40**
- **Effective cost per unit: $200 + $40 = $240**
- **Answer: $240 per unit**

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Manufacturing Cost & NRE

4- Which Cost Is NRE?

A company developing **a smart watch** spends money on:

- Buying microcontrollers for production.
- Designing the PCB.
- Developing the firmware.
- Testing prototypes.
- Assembling finished watches.

Which of these are NRE costs?

Manufacturing Cost & NRE

4- Answer:

- The NRE costs are:
  - **PCB design**
  - **Firmware development**
  - **Prototype testing**
- Buying production components and assembling finished watches are **manufacturing costs**.

Manufacturing Cost & NRE

5- Total Project Cost

A company develops a smart irrigation controller.

- Hardware development = $8,000
- Software development = $12,000
- Prototype testing = $5,000
- Manufacturing cost = $30 per unit
- The company produces **2,000 units**.

Find:

- NRE
- Manufacturing cost
- Total cost

Manufacturing Cost & NRE

5- Answer:

- **1. NRE =** $8,000 + $12,000 + $5,000 = **$25,000**
- **2. Manufacturing cost =** 2,000 × $30 = **$60,000**
- **3. Total cost =** $25,000 + $60,000 = **$85,000**
- **Final answer: $85,000**

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Specification

- The specification is more precise.
  - It serves as the contract between the customer and the architects. As such, the specification must be carefully written so that it accurately reflects the customer's requirements and does so in a way that can be clearly followed during design.
- The **specification** should be understandable enough so that someone can verify that it meets system requirements and overall expectations of the customer.

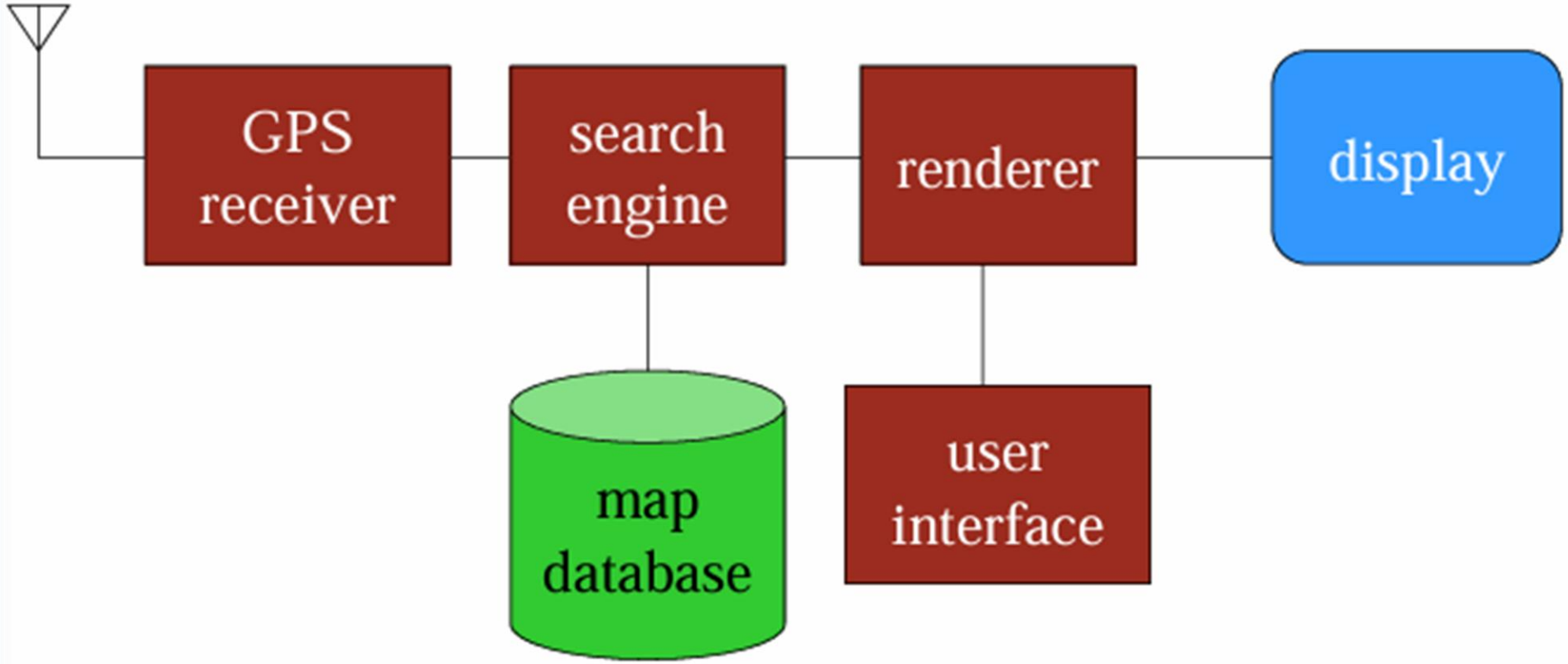
Specification

- A specification of the GPS system would include several components:
  - Data received from the GPS satellite constellation.
  - Map data.
  - User interface.
  - Operations that must be performed to satisfy customer requests.
  - Background actions required to keep the system running, such as operating the GPS receiver.

Architecture design

- What major components go to satisfying the specification?
  - Hardware components:
    - CPUs, peripherals, etc.
  - Software components:
    - major programs and their operations.
    - major data structures
- Evaluate hardware vs. software tradeoffs
- Must take into account functional and non-functional specifications.

GPS moving map block diagram

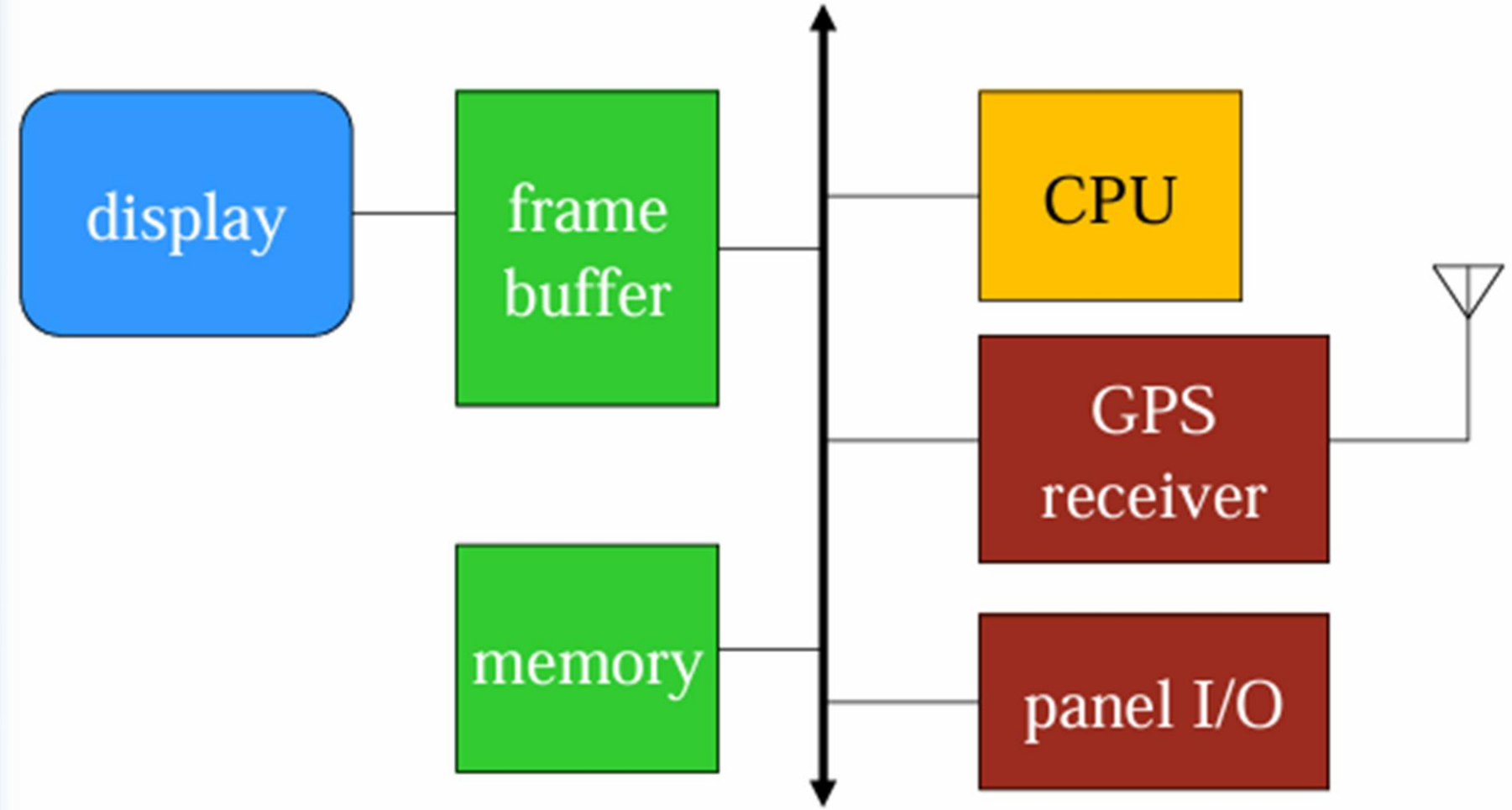


GPS moving map block diagram

The previous diagram shows the components of a GPS-based map system:

- **GPS Receiver**: Determines the current location using satellite signals.
- **Search Engine**: Finds relevant data from the map database based on the location.
- **Renderer**: Converts the data into a visual format for display.
- **Map Database**: Stores map and geographical data like roads and landmarks.
- **User Interface**: Allows the user to interact with the system, like entering destinations.
- **Display**: Shows the map and location to the user.

GPS moving map hardware architecture



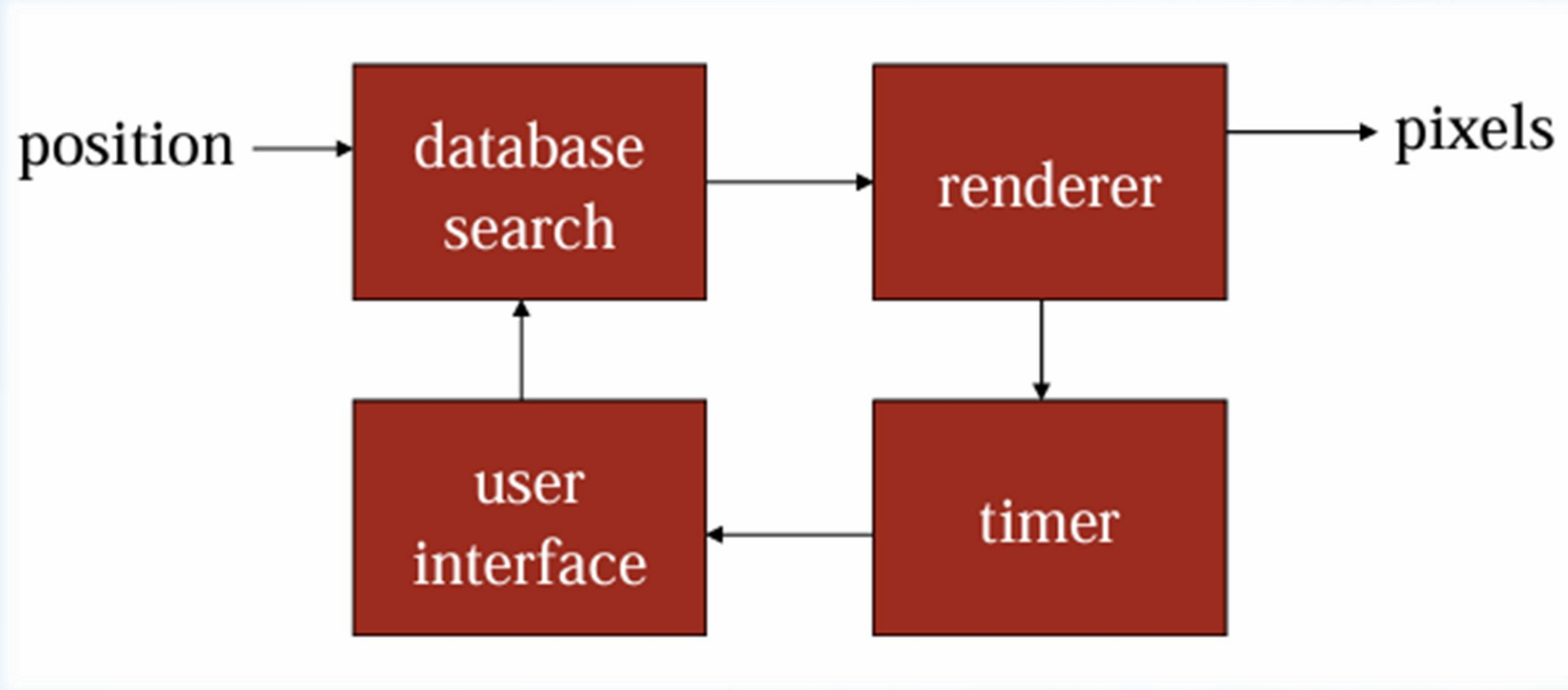
GPS moving map hardware architecture

The previous diagram illustrates the hardware components of a GPS-based display system. Here's a breakdown of the components:

- **Display**: Shows the final output, such as maps or location data.
- **Frame Buffer:** Temporarily stores image data before it's displayed.
- **CPU**: Central processing unit, which manages all the system's tasks, like processing data from the GPS receiver and controlling the display.
- **GPS Receiver:** Receives satellite signals to determine the current location.
- **Memory:** Stores data needed for the system, such as maps and other resources.
- **Panel I/O:** Manages input and output between the system's display panel and the user interface.

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GPS moving map software architecture



GPS moving map software architecture

- **Position**: The current location data, possibly from a GPS receiver.
- **Database Search**: The system searches a database for relevant map data based on the position.
- **Renderer**: Converts the map data into a visual representation, transforming it into pixels that can be displayed.
- **Pixels**: The rendered visual is displayed as pixels on the screen.
- **User Interface**: Allows the user to interact with the system, possibly to change settings or navigate.
- **Timer**: Tracks time intervals, possibly used to update the map or refresh data periodically.

Designing hardware and software components

- Must spend time architecting the system before you start coding or designing circuits.
- Some components are ready-made, some can be modified from existing designs, others must be designed from scratch.

System integration

- Put together the components.
  - Many bugs appear only at this stage.
  - Interfaces must be well designed.
- Have a plan for integrating components to uncover bugs quickly, test as much functionality as early as possible.
- **Test to each specification.**

Microprocessor and Microcontroller

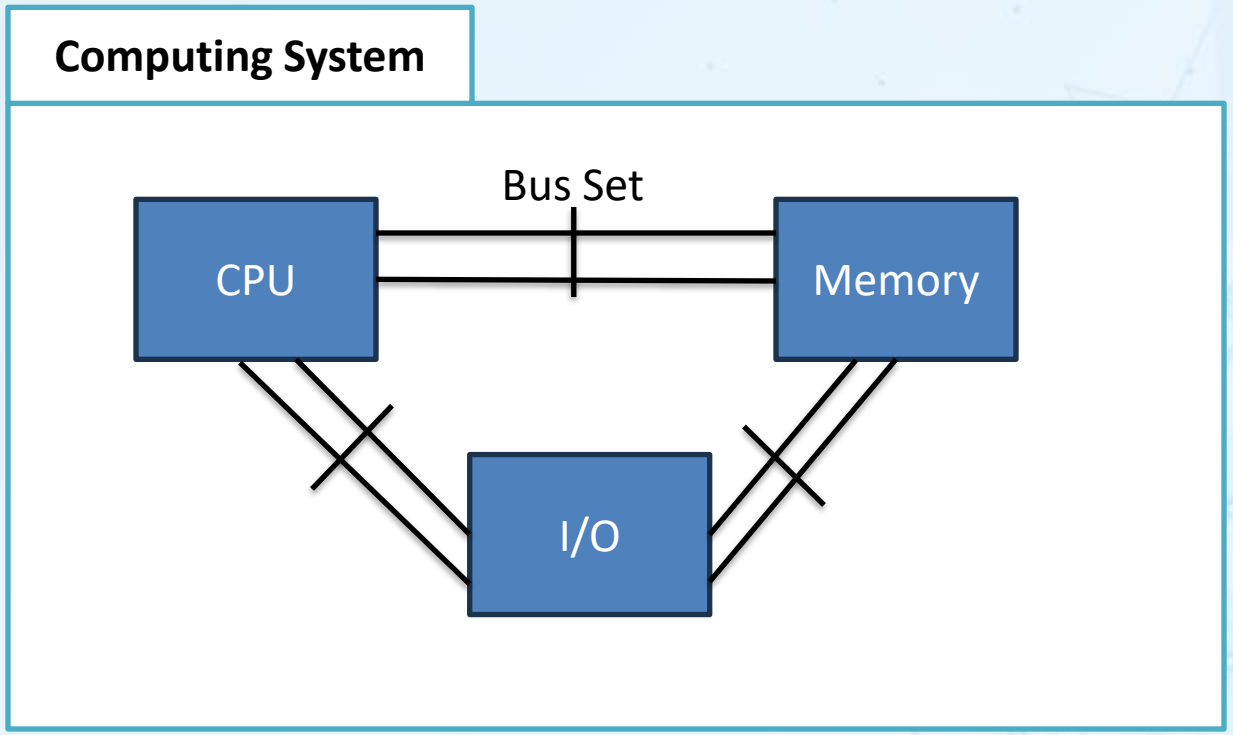
- In general, a microprocessor is a general-purpose device that finds its application in most of the electronic devices.
- It is a dependent unit that requires other chips for its operation.
- Microprocessor is an I.C. which contains many useful functions.
- It requires external memory device to store set of instructions to carry out user-defined task.
- Limited power saving options compared to microcontrollers.

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Microprocessor and Microcontroller

- Microcontroller is a specific purpose device which has specific task for specific device.
- It is an independent device that does not require any other specific chips.
- It is called as microchip which contains the components of microprocessor.
- It has the ability to execute a stored set of instruction to carry out user-defined task.
- Includes lot of power saving features.

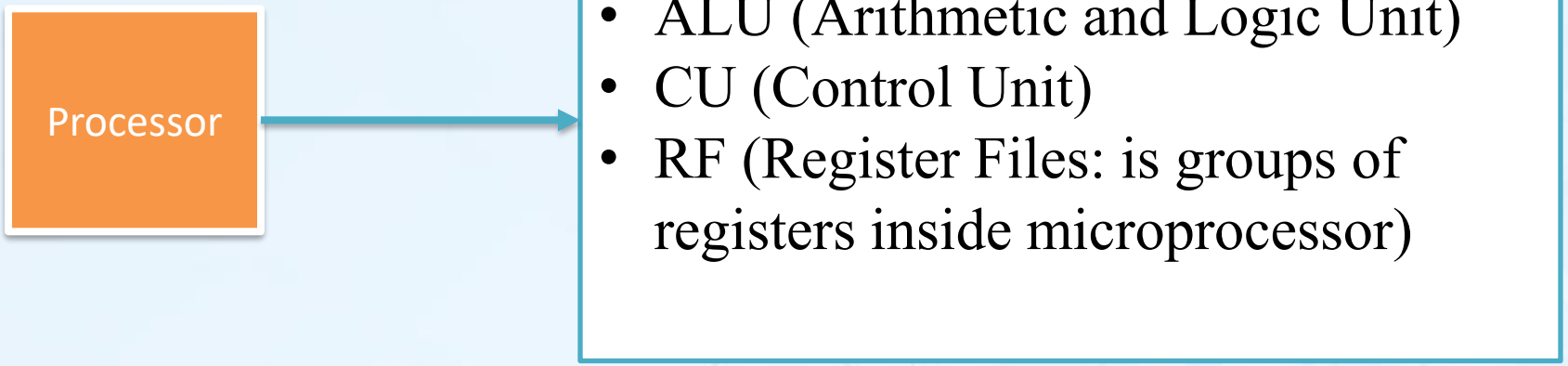
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- CUP is connected to the memory and I/O using Buses.
- Bus set contains of the three main busses (Data bus, Address Bus, and Control Bus)

1. Processor

- Processor consists of three main things:
  - ALU
  - CU
  - RF



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THANK YOU FOR WATCHING

QUESTIONS?

